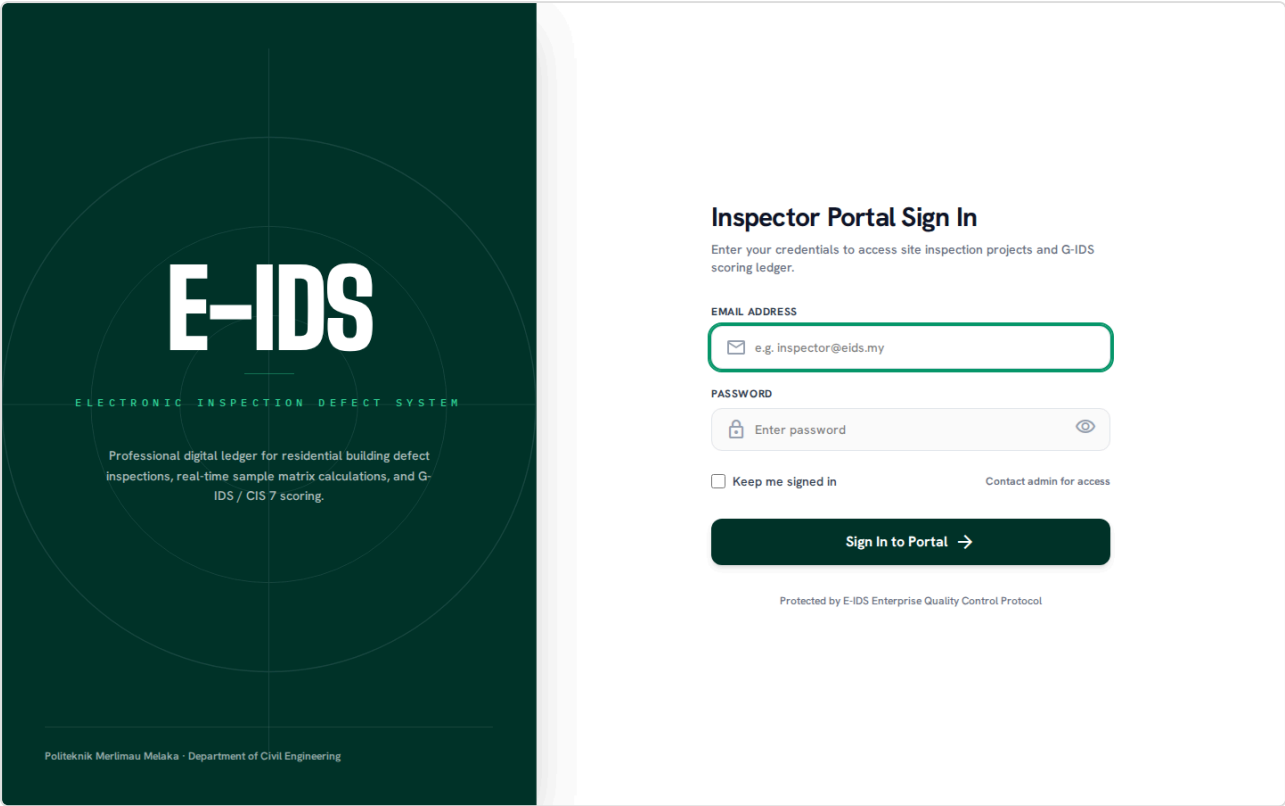


E-IDS v2 — User Manual & Client Guide

Electronic Inspection Defect System (E-IDS v2) *Official Operational Manual for Admins, Site Inspectors, and Contractors*

Updated 2026-08-02 to reflect the simplified 3-role system (Admin, Inspector, Contractor), per-role project/defect visibility, and the 4-state defect lifecycle.

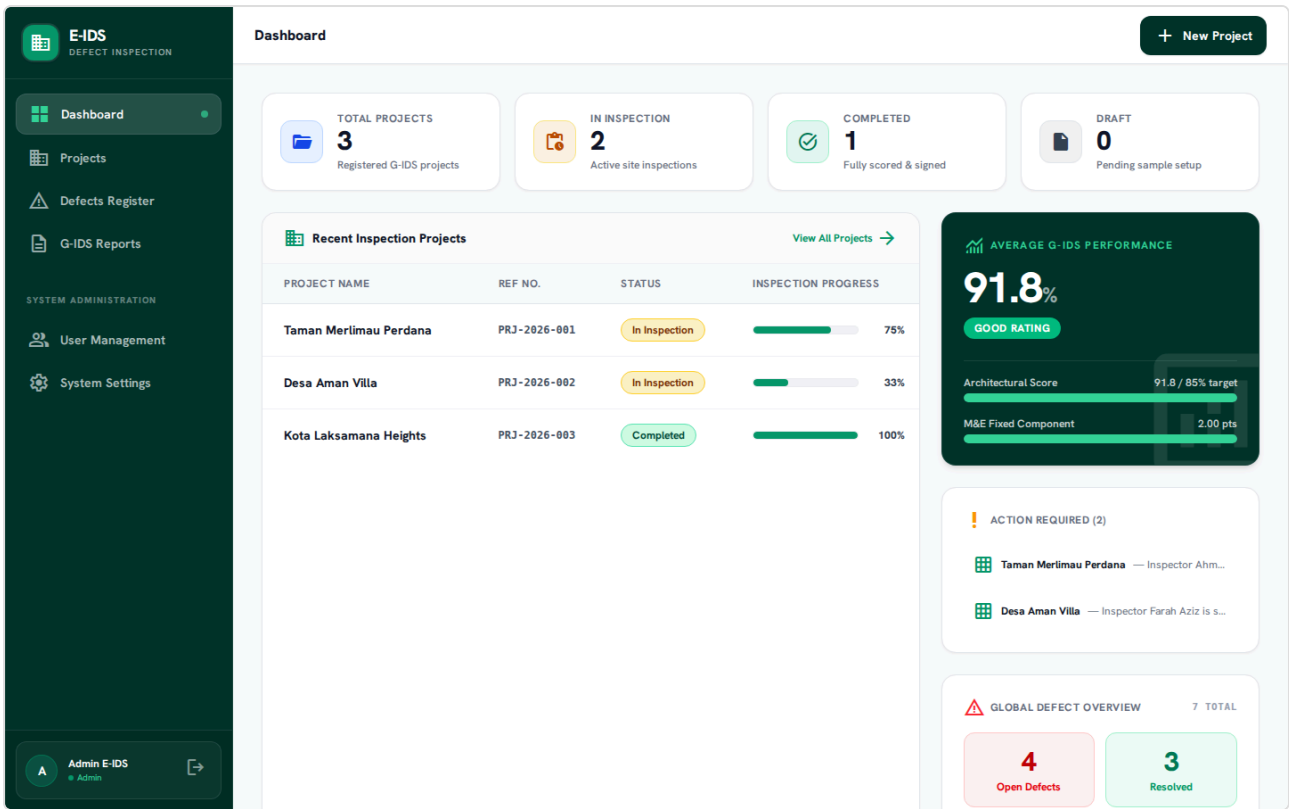


The sign-in screen every role shares — the same login form routes Admin, Inspector, and Contractor to their own scoped view of the system.

1. System Overview

E-IDS v2 is a specialized web-based building defect inspection and scoring platform for the Malaysian construction sector. Built in compliance with G-IDS / CIS 7 quality standards, E-IDS replaces manual paper-based forms with an end-to-end digital workflow:





The Dashboard is the landing page for Admin and Inspector — project totals, the average score across every visible project, and an Action Required list of what still needs attention.

2. User Roles & Permission Matrix

E-IDS v2 supports **three** roles. Each role's view of the system is scoped — Admin sees every project; Inspector and Contractor see only what they're individually assigned to.

Contractor / Developer (Contractor)	Inspector (Inspector)	Admin
No Defects Only — No Dashboard/Projects Access	See Only Their Assigned Projects	User Management & Roles
View Reports in Progress & Solved	Sample Setup & M&E Component Check	Formulas Settings & Tolerances
Control Self-Certify as Resolved	Verify Contractor Reports & Close Defects	See & Manage Every Project
		Register Projects, Assign Inspector/Contractor
		Sign Off Completion & Generate the Signed PDF

Detailed Permissions Table

Action / Capability	Admin	Inspector	Contractor
View Dashboard, Projects & Reports	✓ all	✓ assigned	✗
Download Official PDF Report	✓ all	✗ (view score only)	✗
Register & Edit Projects	✓	✓	✗
Assign the Inspector (assigned_to)	✓	✗ (locked to self)	✗
Assign the Contractor (assigned_contractor_id)	✓	✓	✗
Configure Sample Location Names	✓	✓	✗

Action / Capability	Admin	Inspector	Contractor
Perform 5-Point Checks (PASS / FAIL)	✗ (handed off to Inspector)	✓	✗
Upload Photo Evidence (Spatie Media)	✗	✓	✗
View Defects Register / My Defects	✓ all	✓ assigned projects	✓ own-assigned project only
Create / Manually Log a Defect	✓	✓	✗
Mark Defect IN_PROGRESS ("Start Repair")	✓	✓	✓
Mark IN_PROGRESS → PENDING_VERIFICATION ("Mark Settled")	✓	✓	✓
Confirm RESOLVED or Reject back to IN_PROGRESS	✓	✓	✗
Mark Project as Completed (sign-off)	✓	✗	✗
Manage Users & Role Access	✓	✗	✗
Modify G-IDS Formula Settings	✓	✗	✗
Delete Project (with Modal Confirm)	✓	✗	✗

What "scoped" means in practice: opening a project or defect you don't have visibility into — whether from a list or by typing the URL directly — returns a "Forbidden" page for every role except Admin. This applies uniformly across Projects, the Dashboard, Defects, and Reports.

E-IDS

DEFECT INSPECTION

Dashboard

Projects

Defects Register

G-IDS Reports

SYSTEM ADMINISTRATION

User Management

System Settings

Admin E-IDS

Admin

Dashboard > Projects

+ New Project

Search project name, number or developer...

All Statuses

All Building Types

All Inspectors

Filter

PROJECT	REF NO.	TYPE	STATUS	PROGRESS	SCORE	ACTIONS
Taman Merlimau Perdana Mutiana Development Sdn Bhd	PRJ-2026-001	Terrace	In Inspection	75%	84.8%	View Edit Delete
Desa Aman Villa Impian Setia Sdn Bhd	PRJ-2026-002	Semi-D	In Inspection	33%	92.8%	View Edit Delete
Kota Laksamana Heights Warisan Bina Sdn Bhd	PRJ-2026-003	Bungalow	Completed	100%	97.8%	View Edit Delete

The Projects list — the same page renders different rows for each role: Admin sees every project, Inspector and Contractor only ever see rows they're assigned to.

3. Operational Role Handoff & Project Lifecycle

E-IDS v2 enforces a clear phase handoff between project creation, site inspection, contractor repairs, and final certificate sign-off. Admin's part of the workflow pauses once sample setup is saved — from there, the assigned Inspector and Contractor take over independently, each seeing only what's relevant to them — before Admin returns at the end to sign off and generate the certificate:



E-IDS

DEFECT INSPECTION

Dashboard

Projects

Defects Register

G-IDS Reports

SYSTEM ADMINISTRATION

User Management

System Settings

Admin E-IDS

Admin

Dashboard > Projects > Taman Merlimau Perdana

Edit

WAITING

Inspector Ahmad Firdaus is still inspecting — 3/4 sample units done.

01 Project Details

02 Sample Setup

03 Handled by Inspector

04 Handled by Inspector

G-IDS Score

In Inspection

PRJ-2026-001

Taman Merlimau Perdana

Mertimau, Melaka

84.8% MODERATE RATING

Inspection Progress

3 of 4 sample unit(s) inspected

75%

Inspect Component Grid

Case Ledger

Current: In Inspection

01 Setup & Assignment

Created by Admin E-IDS · Assigned to Ahmad Firdaus

Configure Rooms

02 Field Inspection

Progress 75% · 24 checks assessed

Open Components Grid

03 Defect Rectification

Open 3 · Resolved 1

View Defect Register

04 Final Certificate

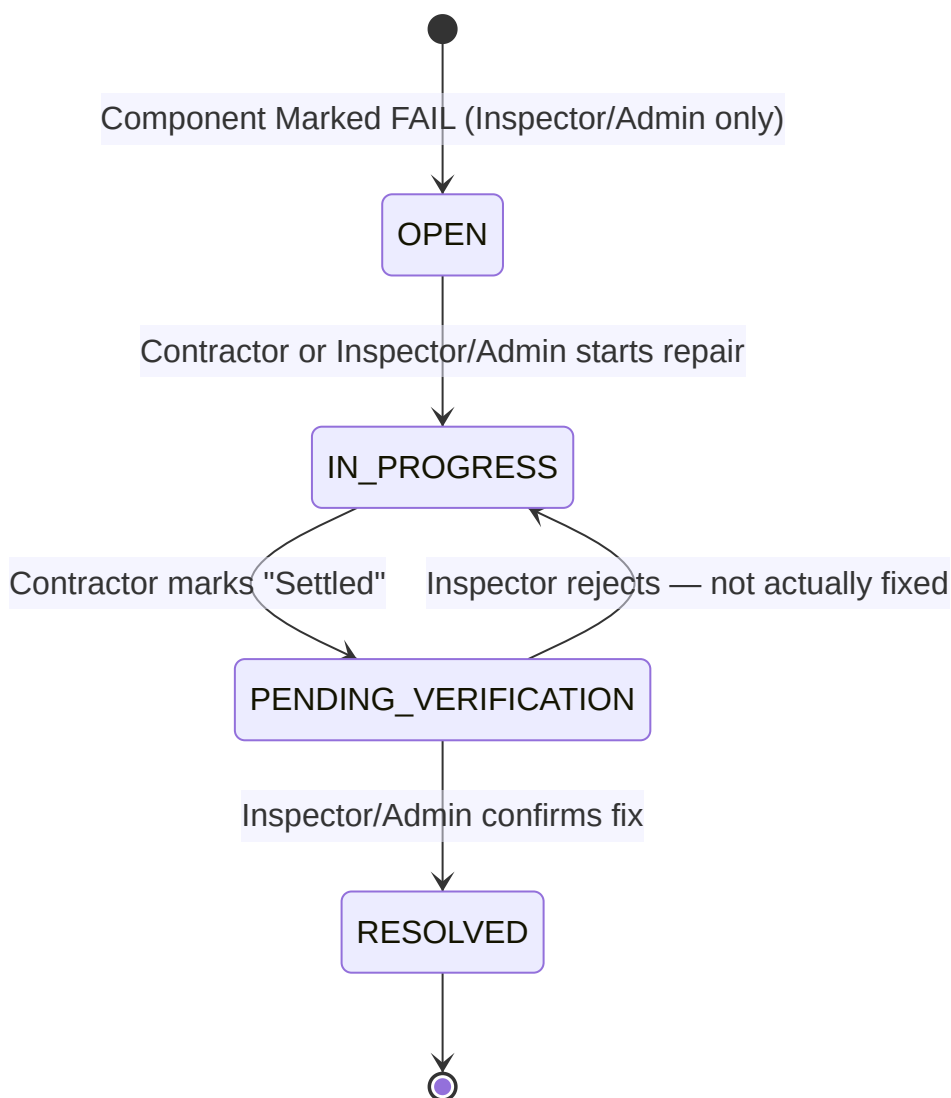
Score 84.8% · Rating MODERATE

Locked

The project page's Case Ledger is the single place this whole handoff is tracked — each row shows who's responsible, the key metric, and one action. Here, Field Inspection is active (green) and Defect Rectification needs attention (red) — Final Certificate stays locked until both clear.

4. Defect Lifecycle & Status Workflow

When a defect is discovered during site inspection, it now follows a **4-stage** lifecycle, splitting the old single "In Progress" phase into a self-reported repair step and a formal verification step:



Defect Status Breakdown

⚠ Important Who updates defect status? The Contractor has their own login and updates status themselves for the repair steps (`IN_PROGRESS` , `PENDING_VERIFICATION`). The final `RESOLVED` confirmation — and the ability to reject a repair back to `IN_PROGRESS` — remains

exclusively with the **Inspector** or **Admin**. A Contractor can never self-certify their own work as fully resolved.

1. ☐ **OPEN (Newly Discovered Defect)**

- **Trigger:** Automatically created when an Inspector (or Admin) marks any of the 5 component criteria (Finishing, Hollow, Levelling, Joint, Crack) as **FAIL** during inspection, or manually logged.
- **Action:** Defect appears in the assigned Contractor's **My Defects** list and the office's **Defects Register**, with severity tag (**Low** , **Medium** , **High**), room location, component code, and photo evidence via Spatie Media Library.

2. ☐ **IN_PROGRESS (Repair Underway)**

- **Trigger:** The Contractor logs in and clicks **Start Repair** on the defect once they begin work on site (e.g., plaster touch-up, tile replacement, re-levelling). An Inspector or Admin can also set this manually if needed.
- **Action:** Status updates in real time so the office knows a repair is genuinely underway, not just reported informally.

3. ☐ **PENDING_VERIFICATION (Awaiting Inspector Sign-off)**

- **Trigger:** The Contractor clicks **Mark Settled** once they believe the repair is complete.
- **Action:** The defect moves into a queue awaiting the Inspector's physical re-check. The Contractor sees a read-only "Awaiting Verification" badge and cannot advance it further themselves.

4. ☐ **RESOLVED (Verified & Closed)**

- **Trigger:** The Inspector (or Admin) conducts a re-inspection of the specified location on site.
- **Action:** Upon verifying the defect has been properly rectified to G-IDS standards, they click **Confirm Resolved**. If the repair is inadequate, they instead click **Reject – Not Fixed**, sending it back to **IN_PROGRESS** for the Contractor to redo.

E-IDS
DEFECT INSPECTION

Dashboard

Projects

Defects Register

G-IDS Reports

SYSTEM ADMINISTRATION

User Management

System Settings

Admin E-IDS

Admin

Dashboard > Defects Register

+ New Defect Log

Search component, location or defect

All Projects

All Severities

All Statuses

Filter

DEFECT / LOCATION	PROJECT	SEVERITY	STATUS	ACTIONS
Internal Wall (Dinding Dalam) Living Room FAIL on A2_WALL (Internal Wall (Dinding Dalam)) at...	PRJ-2026-001 Taman Merimau Perdana	Medium	Open	Start Repair
Window (Tingkap) Living Room FAIL on A5_WINDOW (Window (Tingkap)) at Living...	PRJ-2026-001 Taman Merimau Perdana	Low	In Progress	Mark Settled
Floor (Lantai) Service Area FAIL on A1_FLOOR (Floor (Lantai)) at Service Area.	PRJ-2026-001 Taman Merimau Perdana	High	Pending Verification	Confirm Resolved Reject - Not Fixed
Roof (Bumbung) Passageway FAIL on A7_ROOF (Roof (Bumbung)) at Passageway.	PRJ-2026-001 Taman Merimau Perdana	Medium	Resolved	Reopen
Door (Pintu) Master Bedroom FAIL on A4_DOOR (Door (Pintu)) at Master Bedroom.	PRJ-2026-002 Desa Aman Villa	Low	Open	Start Repair
Ceiling (Siling) Living Room FAIL on A3_CEILING (Ceiling (Siling)) at Living Room.	PRJ-2026-003 Kota Laksamana Heights	Low	Resolved	Reopen

Internal Fixtures

All four states side by side: Open shows Start Repair, In Progress shows Mark Settled, Pending Verification shows Confirm Resolved / Reject – Not Fixed, and Resolved shows Reopen.

5. UI/UX Feature Highlights

□ Role-Aware Guidance

Every project page shows a one-line **"what's next"** banner tailored to the viewer's role — e.g. an Admin sees *"Waiting on Inspector Aiman to begin the Components Grid inspection"*, while an Inspector sees *"2 defect(s) awaiting your verification."* The Dashboard additionally surfaces an **Action Required** widget for Admin/Inspector, listing their own visible projects that still need attention.

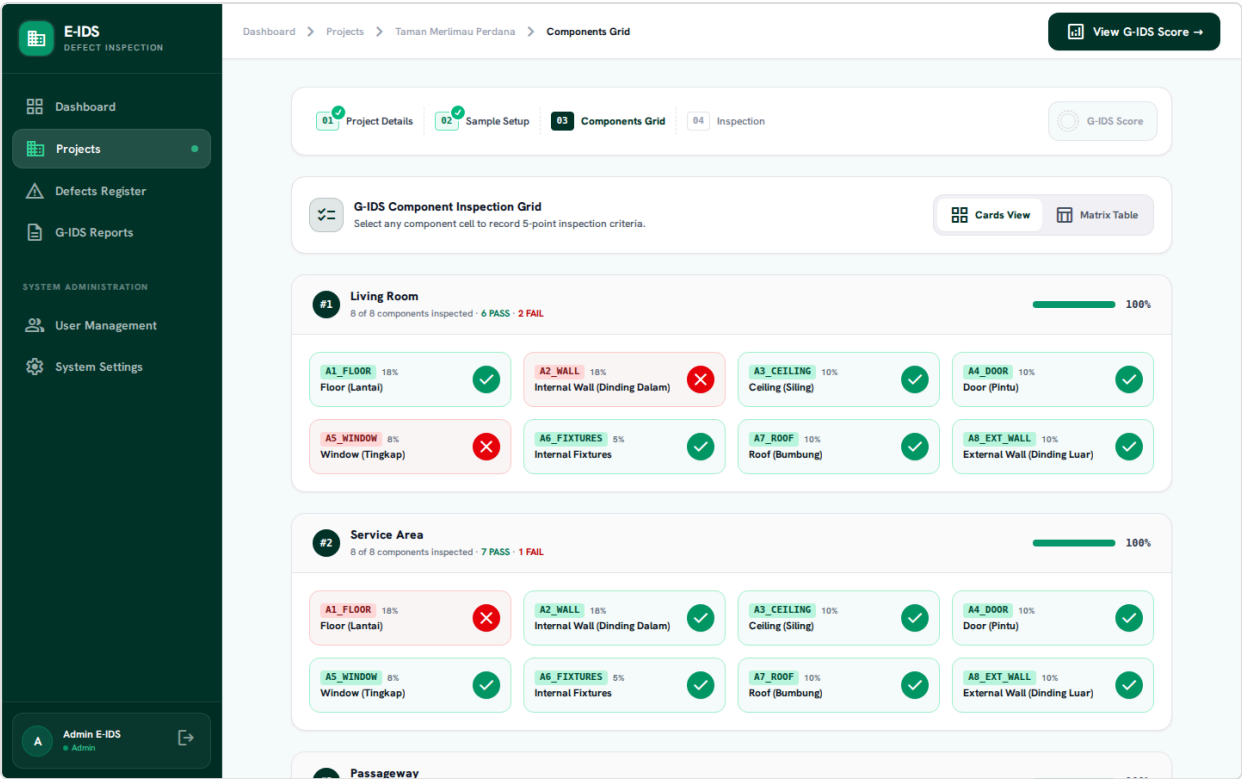
□ Dual View Mode (Components Grid)

To eliminate horizontal side-scrolling on mobile and tablet screens during site walkthroughs, E-IDS v2 provides two view modes on the **Components Grid**:

1. Cards View (Default — Mobile & Touch Optimized):

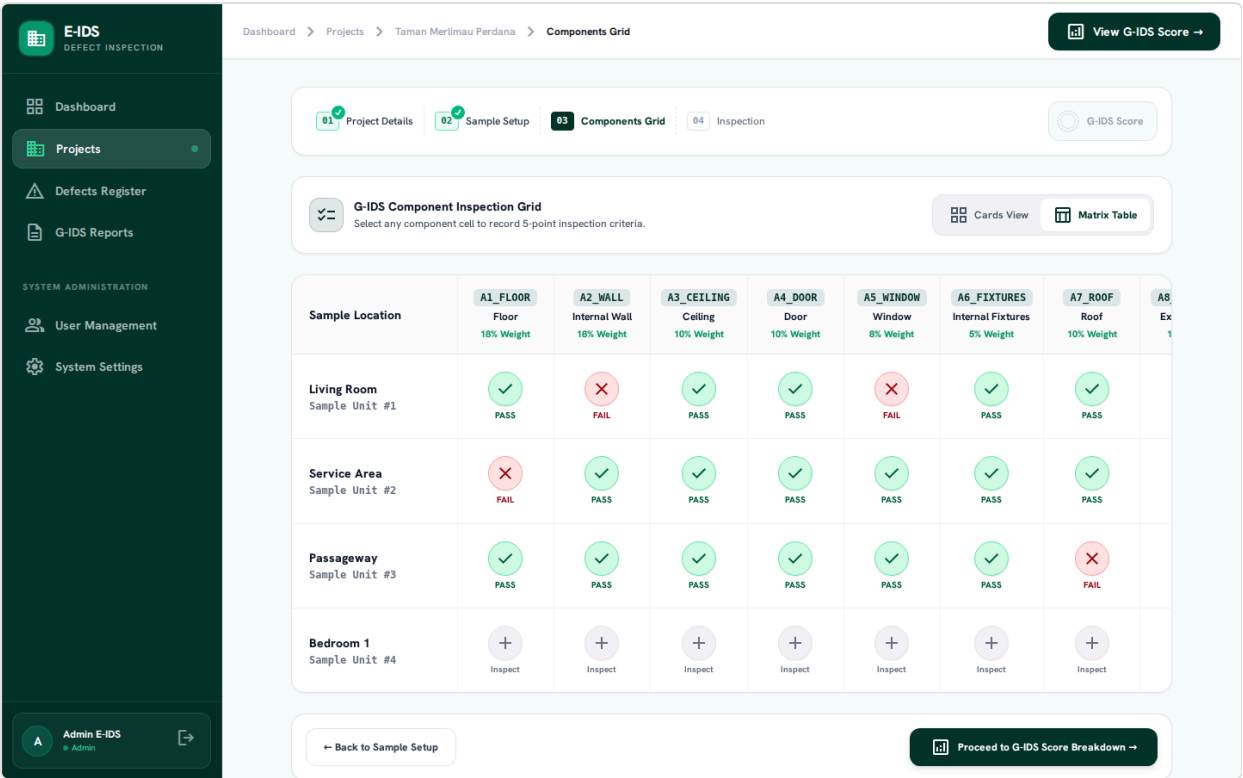
- Sample units are stacked as vertical cards.
- 8 architectural components are rendered in a 2 to 4-column touch grid with high-contrast PASS , FAIL , or PENDING badges.

- Zero side-scrolling required.



2. Matrix Table View (Spreadsheet Mode):

- Traditional table grid for desktop monitors and office review.



□ Spatie Media Library Photo Storage

- Automated thumbnail generation (300 × 300px) for fast table rendering.
- HD preview conversion (800 × 600px) for PDF reports.
- Media items attached to inspection assessments are automatically copied to auto-spawned defect records.

E-IDS

DEFECT INSPECTION

Dashboard

Projects

Defects Register

G-IDS Reports

A

Ahmad Firdaus

Inspector

Dashboard > Taman Merlimau P... > Grid > Living Room / A1_FLOOR

01 Project Details

02 Sample Setup

03 Components Grid

04 Inspection

G-IDS Score

Existing assessment loaded for A1_FLOOR

Last saved 59 minutes ago

TAMAN MERLIMAU PERDANA · COMPONENT 1 OF 8

Floor (Lantai) (A1_FLOOR)

Living Room · Sample Unit #1

ARCHITECTURAL WEIGHT

18%

CALCULATED COMPONENT RESULT

PASS

✓ Inspection Criteria Checklist (5 Checks)

44PX TOUCH TARGET

1. Finishing

Surface finishing quality, alignment & plastering uniform

PASS

FAIL

2. Hollow

Tapping test for hollow sound underneath plastering or tiles

PASS

FAIL

3. Levelling (mm)

Max allowable tolerance: 3mm

1.20

mm

PASS

The 5-point assessment form for one component — Finishing, Hollow, Levelling, Joint/Gap, and Crack, with the calculated PASS/FAIL result and photo evidence upload.

Modal Confirmation Dialogs

- All destructive actions (Deleting Projects, Deleting Defects, Deleting User Accounts) are protected by an Alpine.js modal dialog (`x-modal-confirm`) requiring explicit confirmation.

6. G-IDS Scoring Framework

The total G-IDS score (S_{total}) is calculated using native CIS 7 rules:

$$S_{total} = S_{arch} + S_{ME} + S_{external}$$

- **Architectural Subtotal (S_{arch}):** Sum of 8 weighted component pass rates (S_{comp}):

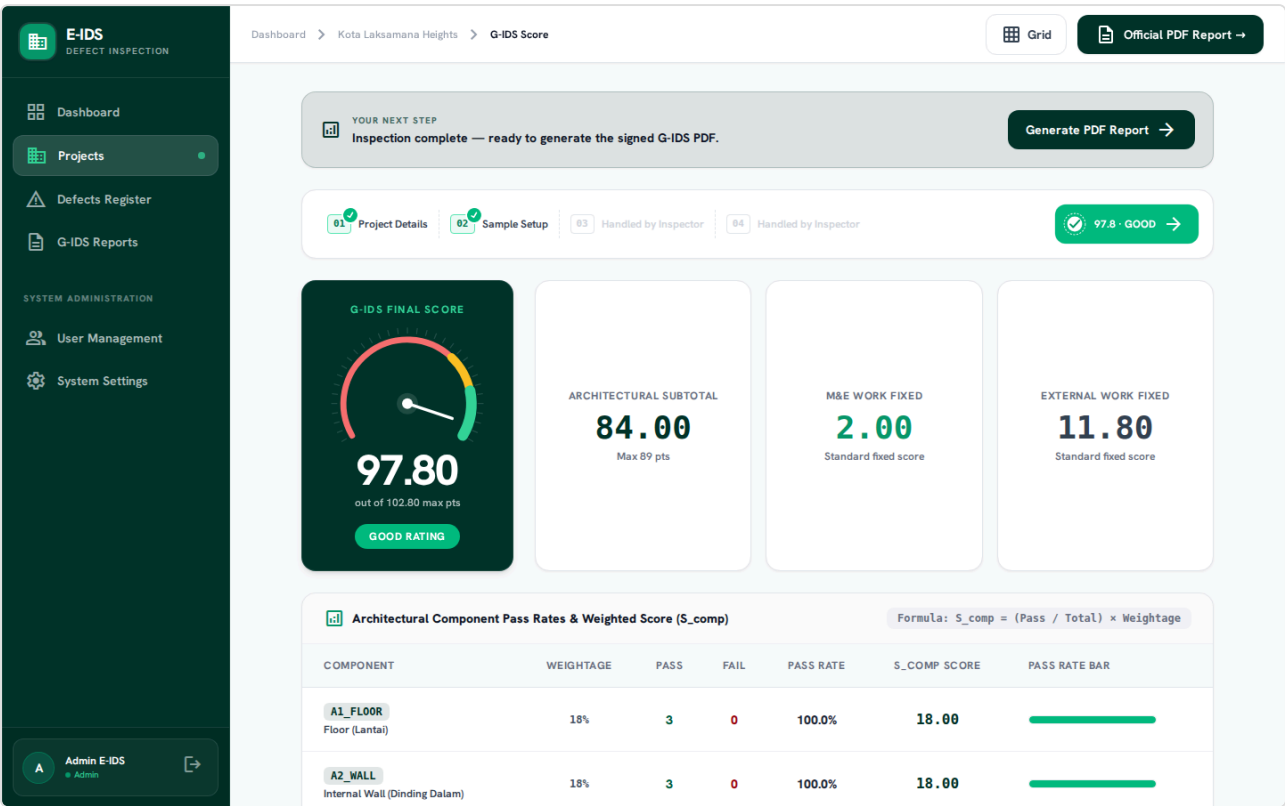
$$S_{\text{comp}} = \left(\frac{\text{Pass Count}}{\text{Total Assessed}} \right) \times \text{Weightage \%}$$

- **M&E Fixed Score (S_{ME}):** 2.00 pts
- **External Work Fixed Score (S_{external}):** 11.80 pts

Rating Thresholds

Note

- ☐ **GOOD RATING:** Total score ≥ 85.00 pts
- ☐ **MODERATE RATING:** Total score ≥ 70.00 pts and < 85.00 pts
- ☐ **WEAK RATING:** Total score < 70.00 pts



The G-IDS Score page: the gauge's needle position and red/amber/green zones map directly to the rating thresholds above, with the full component-by-component breakdown below.

7. Step-by-Step Operating Instructions

Step 1: Project Registration & Team Assignment (Admin)

1. Navigate to **Projects** → Click **+ New Project**.
2. Enter Project Ref No., Project Name, Developer, Contractor, Building Type, and Total Units.
3. Enter Gross Floor Area (GFA m²). The system auto-calculates sample units ($N = \lceil \text{GFA} \div 60 \rceil$).
4. Select the **Assigned Inspector** and **Assigned Contractor** from the dropdowns. (An Inspector creating a project instead is auto-assigned to themselves, and can still choose the Contractor — only Admin can assign the Inspector field.)
5. Click **Create Project**.

The screenshot shows the 'New Project' form in the E-IDS system. The form is divided into two main sections: 'Project Specifications & Metadata' and 'Sample Unit Calculation'.

Project Specifications & Metadata

- PROJECT REFERENCE NO. ***: e.g. PRJ-2026-001
- PROJECT NAME ***: e.g. Taman Merlimau Perdana
- SITE LOCATION / ADDRESS**: e.g. Merlimau, Melaka
- DEVELOPER NAME ***: Developer company name
- CONTRACTOR NAME ***: Contractor company name
- BUILDING TYPE ***: Select building type...
- INITIAL STATUS ***: In Inspection
- ASSIGNED INSPECTOR**: Unassigned (Open for all inspectors)
- ASSIGNED CONTRACTOR**: Unassigned
- TOTAL PROJECT UNITS ***: 1

Sample Unit Calculation

Formula: $N = \max(1, \lceil \text{GFA} / 60 \rceil$)

GROSS FLOOR AREA (GFA M²) *: e.g. 210.00

CALCULATED SAMPLE UNITS (N): 0

Step 2: Sample Location Setup (Admin / Inspector)

1. System automatically redirects to **Sample Setup**.
2. Select default location names (e.g. *Master Bedroom*, *Living Room*, *Kitchen*) or enter custom room names.
3. Click **Save Locations**. For Admin, this pauses their part of setup — you're returned to the Project Overview page. The project now appears under the assigned Inspector's **My Assigned Projects**, ready for on-site inspection. (An Inspector doing their own setup instead continues straight into the Components Grid.)

E-IDS

DEFECT INSPECTION

Dashboard

Projects

Defects Register

G-IDS Reports

SYSTEM ADMINISTRATION

User Management

System Settings

Admin E-IDS

Admin

Dashboard > Projects > Taman Merlimau Perdana > Configure Samples

01 Project Details

02 Sample Setup

03 Components Grid

04 Inspection

G-IDS Score

Sample Units Configuration

Formula: $N = \max(1, \text{ceil}(\text{GFA} \times 60)) = 4 \text{ sample units}$. Assign a room or location name to each sample unit. The assigned inspector will perform the component inspection separately.

Sample Units Location Assignment (4)

GFA: 210.00 m²

#1	Living Room	Inspected
#2	Service Area	Inspected
#3	Passageway	Inspected
#4	Bedroom 1	Pending

Architectural Components Framework (8 Components)

A1_FLOOR	Floor (Lantai)	18% Weightage
A2_WALL	Internal Wall (Dinding Dalam)	18% Weightage

Step 3: Component Inspection Matrix Grid (Inspector)

- Open **Components Grid** (defaults to mobile-friendly **Cards View**) from **My Assigned Projects**.
- Click any component tile (**D1** , **D2** , etc.) to open the **5-Point Assessment Form**:
 - Finishing**: PASS / FAIL
 - Hollow**: PASS / FAIL
 - Levelling (mm)**: Enter measured mm (auto-checked against $\leq 3.0\text{mm}$ tolerance)
 - Joint/Gap (mm)**: Enter measured mm (auto-checked against $\leq 1.0\text{mm}$ tolerance)
 - Crack**: PASS / FAIL
- Upload photo evidence if defects exist.
- Click **Save & Next** to advance through component slots smoothly. A **FAIL** automatically opens a defect in the Contractor's queue.

Step 4: Repair & Verification (Contractor, then Inspector)

- Contractor** logs in and lands directly on **My Defects** — a list scoped to their one assigned project.
- Click **Start Repair** on an **OPEN** defect once work begins on site (moves it to **IN_PROGRESS**).
- Click **Mark Settled** once the repair is complete (moves it to **PENDING_VERIFICATION**).
- Inspector** opens the **Defects Register**, filters to **Pending Verification** , and re-inspects the location on site.

- Inspector clicks **Confirm Resolved** if the fix holds up, or **Reject – Not Fixed** to send it back to `IN_PROGRESS` for another attempt.

Step 5: Score Review (Admin / Inspector)

- Once every sample unit is inspected, the certification-seal button in the step bar unlocks — click it, or **G-IDS Score**, to review the final score, rating, and architectural component pass rates.
- If any defect is still `OPEN`, `IN_PROGRESS`, or `PENDING_VERIFICATION`, the score page and PDF report stay locked with an explanatory message until every defect reaches `RESOLVED`.

Step 6: Completion Sign-off & PDF Export (Admin only)

- Once inspection is 100% done and every defect is `RESOLVED`, the project page's **Final Certificate** panel shows a **Mark as Completed** button.
- Click it to formally close out the project (status becomes **Completed**) — this is a deliberate action; nothing marks a project Completed automatically.
- Click **Generate Official PDF Report** to view or print the formal signed G-IDS Inspection Certificate.

Dashboard > Kota Laksamana Heights > G-IDS Report

Export Signed PDF Certificate →

FORMAL G-IDS INSPECTION CERTIFICATE

Kota Laksamana Heights

PRJ-2026-003 · Bungalow
Bandar Hilir, Melaka

FINAL G-IDS SCORE
97.80
GOOD RATING

84.00 Architectural Subtotal	2.00 M&E Work (Fixed)	11.80 External Work (Fixed)	2 Total Defects Logged
--	---------------------------------	---------------------------------------	----------------------------------

Project Information & Audit Metadata

DEVELOPER Warisan Bina Sdn Bhd	CONTRACTOR Kukuh Sentosa Construction Sdn Bhd	TOTAL UNITS 12 units
GROSS FLOOR AREA (GFA) 180.00 m²	LEAD INSPECTOR Admin E-IDS	AUDIT COMPLETION DATE 02 Aug 2026

Architectural Component Score Breakdown

$S_{comp} = (Pass / Total) \times Weightage$

COMPONENT	WEIGHTAGE	PASS	FAIL	PASS RATE	S_COMP SCORE
A1_FLOOR Floor (Lantai)	18%	3	0	100.0%	18.00
A2_WALL Internal Wall (Dinding Dalam)	18%	3	0	100.0%	18.00
A3_CEILING Ceiling (Siling)	10%	2	1	66.7%	6.67

Admin E-IDS
Admin

The signed certificate — only reachable once inspection is 100% done and every defect is `RESOLVED`. Its "Export Signed PDF Certificate" button in the top bar produces the downloadable file.

8. Demo / Testing Accounts

Running `docker exec -w /var/www eidsv2 php artisan migrate:fresh --seed` loads a ready-to-test dataset: **3 projects and 5 accounts covering all 3 roles**, split into an "A" set (Project 1) and a "B" set (Project 2), plus a third fully-completed project so every phase of the lifecycle — including the ones that only show up at the very end — has something real to look at.

Role	Account	Password	Scope
Admin	admin@eids.gov.my	password	Sees everything; created all three projects
Inspector	inspector.a@eids.gov.my	password	Assigned to Project 1 and Project 3
Inspector	inspector.b@eids.gov.my	password	Assigned to Project 2 only
Contractor	contractor.a@eids.gov.my	password	Assigned to Project 1 only — lands on My Defects
Contractor	contractor.b@eids.gov.my	password	Assigned to Project 2 and Project 3 — lands on My Defects

Project 1 — "Taman Merlimau Perdana" (PRJ-2026-001): 4 sample rooms, 3 fully inspected + 1 left pending (so you can see the "Continue — 3/4 done" banner). One defect seeded in **every lifecycle state** so every action button can be tested immediately:

Defect	Location / Component	Status	What to test
1	Living Room / Internal Wall	OPEN	Contractor A sees Start Repair
2	Living Room / Window	IN_PROGRESS	Contractor A sees Mark Settled
3	Service Area / Floor	PENDING_VERIFICATION	Inspector A sees Confirm Resolved / Reject ; Contractor A sees a read-only "Awaiting Verification" badge
4	Passageway / Roof	RESOLVED	Inspector A sees Reopen

Score: 84.80 — **MODERATE** (just under the GOOD threshold, useful for demonstrating the rating boundary).

Project 2 — "Desa Aman Villa" (PRJ-2026-002): smaller, minimal setup — one **OPEN** defect only. Its main purpose is proving Contractor B / Inspector B never see Project 1's data. Score: 92.80 — **GOOD**.

Project 3 — "Kota Laksamana Heights" (PRJ-2026-003): the only seeded project that's **fully signed off** (status **Completed**) — 100% inspected, both of its defects already **RESOLVED**. Use this one to see what everything else in this manual only *describes*: the unlocked certification-seal button in the stepper, the Case Ledger with all four rows stamped complete, and the actual **Official PDF Certificate** page. Score: 97.80 — **GOOD**.